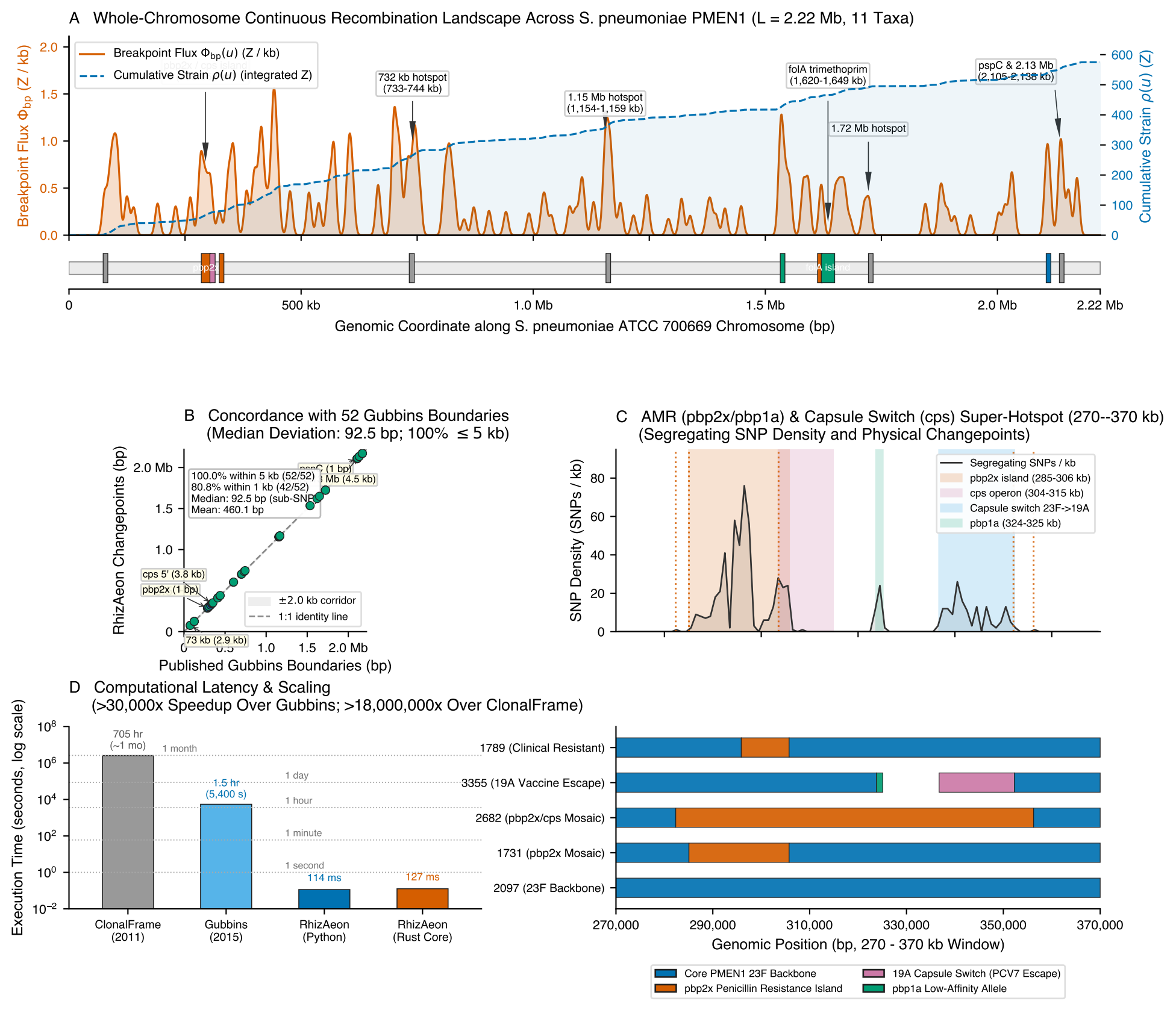

A Whole-Chromosome Continuous Recombination Landscape Across S. pneumoniae PMEN1 (L = 2.22 Mb, 11 Taxa)
Breakpoint Flux Φbp(u) (Z / kb)
Cumulative Strain ρ(u) (integrated Z)
732 kb hotspot (733-744 kb)
1.15 Mb hotspot (1,154-1,159 kb)
folA trimethoprim (1,620-1,649 kb)
1.72 Mb hotspot
pspC & 2.13 Mb (2,105-2,138 kb)
Breakpoint Flux Φbp (Z / kb)
Cumulative Strain ρ(u) (Z)
pbp2x
folA island
0
500 kb
1.0 Mb
1.5 Mb
2.0 Mb
2.22 Mb
Genomic Coordinate along S. pneumoniae ATCC 700669 Chromosome (bp)
B Concordance with 52 Gubbins Boundaries
(Median Deviation: 92.5 bp; 100% ≤ 5 kb)
100.0% within 5 kb (52/52)
80.8% within 1 kb (42/52)
Median: 92.5 bp (sub-SNP)
Mean: 460.1 bp
pspC (1 bp)
Mb (4.5 kb)
cps 5' (3.8 kb)
pbp2x (1 bp)
73 kb (2.9 kb)
±2.0 kb corridor
1:1 identity line
RhizAeon Changepoints (bp)
Published Gubbins Boundaries (bp)
C AMR (pbp2x/pbp1a) & Capsule Switch (cps) Super-Hotspot (270--370 kb)
(Segregating SNP Density and Physical Changepoints)
Segregating SNPs / kb
pbp2x island (285-306 kb)
cps operon (304-315 kb)
Capsule switch 23F->19A
pbp1a (324-325 kb)
SNP Density (SNPs / kb)
D Computational Latency & Scaling
(>30,000x Speedup Over Gubbins; >18,000,000x Over ClonalFrame)
Execution Time (seconds, log scale)
705 hr (~1 mo)
1 month
1 day
1.5 hr (5,400 s)
1 hour
1 minute
1 second
114 ms
127 ms
ClonalFrame (2011)
Gubbins (2015)
RhizAeon (Python)
RhizAeon (Rust Core)
1789 (Clinical Resistant)
3355 (19A Vaccine Escape)
2682 (pbp2x/cps Mosaic)
1731 (pbp2x Mosaic)
2097 (23F Backbone)
270,000
290,000
310,000
330,000
350,000
370,000
Genomic Position (bp, 270 - 370 kb Window)
Core PMEN1 23F Backbone
pbp2x Penicillin Resistance Island
19A Capsule Switch (PCV7 Escape)
pbp1a Low-Affinity Allele